

CHAPTER -2

VIDEO DISPLAY DEVICES:

Display devices are output devices used to show information generated by a computer. The most commonly used display device in computer graphics is the video monitor. It displays text, images, and graphics on the screen for users to view. Traditionally, many video monitors operated using Cathode Ray Tube (CRT) technology.

How the Interactive Graphics display works:

A modern interactive graphics display is simple in design and consists of three main components:

- 1) A **digital memory or frame buffer**, Stores the image as a matrix of pixel intensity values.
- 2) A **monitor**: Displays the image on the screen.
- 3) A **display controller**, which is a simple interface that transfers the image data from the frame buffer to the monitor

The image is stored in the frame buffer as a collection of binary numbers representing a rectangular arrangement of pixels (picture elements). A pixel is the smallest unit that can be displayed on the screen.

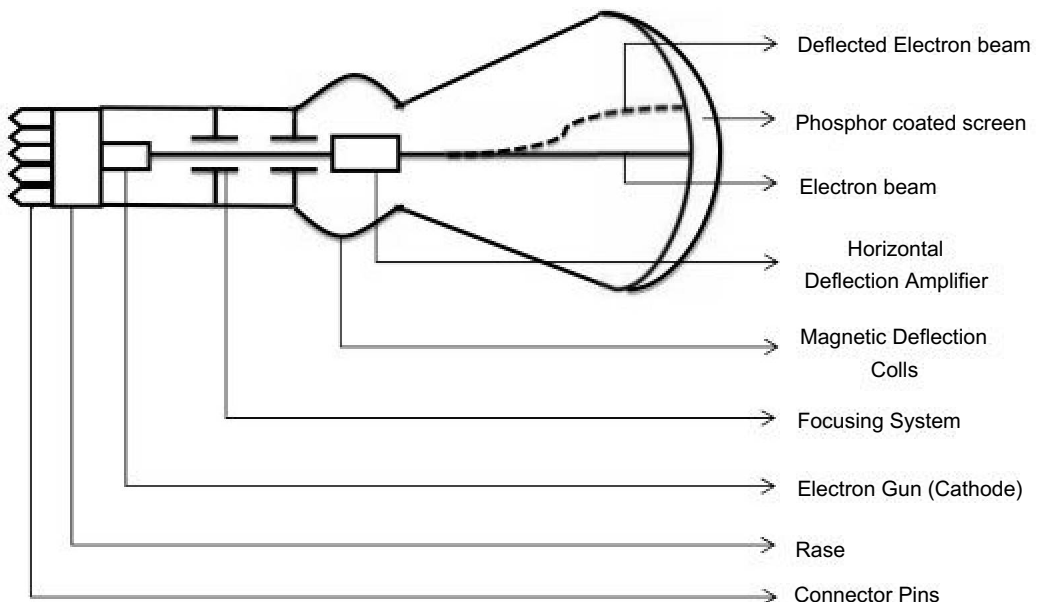
In a simple black-and-white display, a pixel value of 0 represents black, while a value of 1 represents white. The display controller reads the pixel data from the frame buffer and converts it into video signals, which are then sent to the monitor to produce the image. To change the displayed picture, the contents of the frame buffer are modified, and the updated image is shown on the screen.

CATHODE-RAY-TUBES:

A **Cathode-Ray Tube (CRT)** is a video display device, whose design depends on the operation of most video.

BASIC DESIGN OF A MAGNETIC DEFLECTION:

A Cathode- Ray Tube (CRT) is a video display device used in traditional monitors and televisions. In a CRT (Cathode Ray Tube), an electron gun produces a stream of electrons. This stream passes through focusing and deflection systems, which control and direct the electron beam to the correct position on the screen on a phosphor- coated screen. When the electron beam strikes the phosphor coating, it produces a small spot of light, forming images on the screen. The light produced by the phosphor fades very quickly. Therefore, to keep the image visible, the electron beam must repeatedly redraw the picture by moving over the same points on the screen many times per second. This process is known as refreshing, and the number of times the screen is redrawn each second is called the refresh rate. A higher refresh rate produces a more stable and flicker-free display.



Operation of an Electron Gun with an Accelerating Anode

An electron gun is the main component of a Cathode Ray Tube (CRT) that generates, accelerates, and focuses a beam of electrons onto the phosphor-coated screen to produce images. The accelerating anode plays a vital role by increasing the speed of the electrons, enabling them to produce a bright and sharp display.

Step-by-Step Operation

1. Heating the Cathode

- The cathode is heated by a filament (heater).
- Due to thermionic emission, the heated cathode releases a large number of free electrons into the vacuum inside the CRT.

2. Controlling the Electron Beam

- The emitted electrons pass through the control grid, which is maintained at a negative potential with respect to the cathode.
- By varying the grid voltage, the number of electrons passing through can be controlled.
- This controls the intensity (brightness) of the image displayed on the screen.

3. Acceleration by the Accelerating Anode

- After passing through the control grid, the electrons are attracted toward the accelerating anode, which is kept at a high positive voltage.
- The accelerating anode increases the kinetic energy and velocity of the electrons.
- As a result, the electron beam becomes fast enough to produce a bright spot when it strikes the phosphor screen.

4. Focusing the Electron Beam

- The accelerated electrons then pass through the focusing anode.
- The focusing anode acts like an electrostatic lens, concentrating the electrons into a thin, narrow beam.
- Proper focusing ensures that the displayed image is sharp and clear.

5. Deflecting the Electron Beam

- The focused electron beam passes through the horizontal and vertical deflection plates (or magnetic deflection coils).
- These deflection systems guide the beam to different positions on the screen according to the input signals.

6. Producing the Image

- Finally, the high-speed electron beam strikes the phosphor-coated screen.
- The phosphor emits visible light where the electrons hit, producing a bright spot.
- By continuously scanning the beam across the screen, complete images or graphics are formed.

CRT is dependent on the type of phosphor, the intensity to be displayed and the focusing and the deflection system.

“Aspect Ratio” is the property of video monitors. This number gives the ratio of vertical points to horizontal points necessary to produce equal length lines in both directions on the screen. An aspect ratio of 3/4 means that a vertical line plotted with 3 points has the same length as a horizontal line plotted with 4 points.

COLOR CRT MONITORS:

A Color CRT monitor displays color images by using different types of phosphor materials that produce different colors of light. By mixing these colored lights together, many different colors can be shown on the screen.

There are two main methods used to produce colors in a CRT monitor:

I. BEAM PENETRATION METHOD:

The Beam Penetration Method is used in Random-Scan Monitors to display color images. In this method, the inside of the CRT screen is coated with two phosphor layers, usually red and green. The color shown on the screen depends on how deeply the electron beam enters these layers. A slow electron beam excites only the outer red layer and produces red color. A fast electron beam passes through the red layer and excites the inner green layer, producing green color. At medium beam speeds, both red and green layers are excited, creating orange and yellow colors. The beam speed, and therefore the color displayed, is controlled by the accelerating voltage of the electron beam.

II. SHADOW MASK METHOD:

The Shadow Mask Method is commonly used in Raster- Scan Displays because it can produce many more colors than the Beam Penetration Method. In this method, the screen contains three types of phosphor dots: Red (R), Green (G), and Blue (B). The CRT has three electron guns, one for each color, and a shadow mask placed behind the screen.

The electron beams from the three guns pass through tiny holes in the shadow mask and strike the correct red, green, and blue phosphor dots. By combining different amounts of red, green, and blue light, a wide range of colors can be produced on the screen.

Composite Monitors are modified TV sets that display combined picture information without requiring a broadcast signal.

Most color CRT graphics systems use RGB monitors, which are based on the Shadow Mask Method. These monitors receive color intensity values directly from the computer. An RGB system with 24 bits per pixel is called a True Color System or Full-Color System because it can display millions of different colors.

FLAT PANEL DISPLAY:

A Flat Panel Display is a type of display device that is thinner, lighter, and uses less power than a CRT monitor. These displays are commonly used in calculators, laptops, portable video games, small TVs, and portable monitors. Flat panel displays are mainly of two types: Emissive Displays and Non-Emissive Displays.

Most flat panel displays use Liquid Crystal Display (LCD) technology. An LCD consists of two glass plates with liquid crystal material placed between them. Horizontal conductors are placed on one glass plate and vertical conductors on the other. The intersection of a row and a column forms a pixel.

In a Passive Matrix LCD, light normally passes through the liquid crystal and makes the pixel visible on the screen. When a voltage is applied to a row and column conductor, the liquid crystal molecules align and block the light. As a result, the pixel turns off.

A more advanced LCD uses a Thin Film Transistor (TFT) at each pixel location. These transistors control the voltage of each pixel and prevent charge leakage, resulting in better image quality. Such displays are called Active Matrix Displays

I. EMISSIVE DISPLAY:

The Emissive Displays (Emitters) are the devices that convert electrical energy into light. E.g.: Plasma Panels, Thin Film Electroluminescent Displays and Light Emitting Diodes are example of emissive displays.

a. PLASMA PANEL:

Plasma Panel, also called a Gas-Discharge Display, contains a gas mixture (usually including neon) between two glass plates. Horizontal and vertical conductors are placed on the glass plates. When voltage is applied to intersecting conductors, the gas at that point glows and produces light. Earlier plasma panels could display only one color, but modern plasma displays can show colors and grayscale images.

b. Thin Film ELECTROLUMINESCENT DISPLAYS:

A Thin-Film Electroluminescent Display is similar to a plasma panel, but it uses a phosphor material instead of gas between the glass plates. When a high voltage is applied to crossing electrodes, the phosphor emits light at that point. These displays require more power than plasma panels, and producing good color and grayscale images is more difficult.

c. LIGHT-EMITTING DIODE (LED):

In an LED Display, a matrix of LEDs is arranged to form the pixels of the screen. The image information is stored in a refresh buffer, and the LEDs light up to display the picture.

II. NON-EMISSIVE DISPLAY:

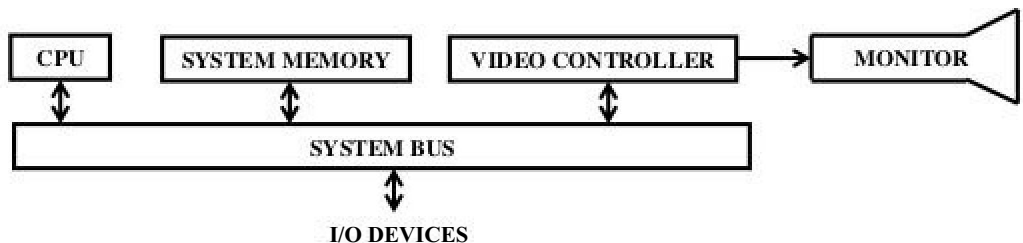
Non-Emissive Displays do not produce their own light. Instead, they use light from an external source or a built-in backlight and control that light to create images. The most common example is the Liquid Crystal Display (LCD).

a. LIQUID CRYSTAL DISPLAY (LCD):

An LCD uses liquid crystal material to control the passage of light. The liquid crystals can either allow light to pass through or block it, creating images on the screen. LCDs are widely used in calculators, laptops, computer monitors, mobile phones, and televisions because they are thin, lightweight, and energy efficient.

RASTER-SCAN SYSTEM:

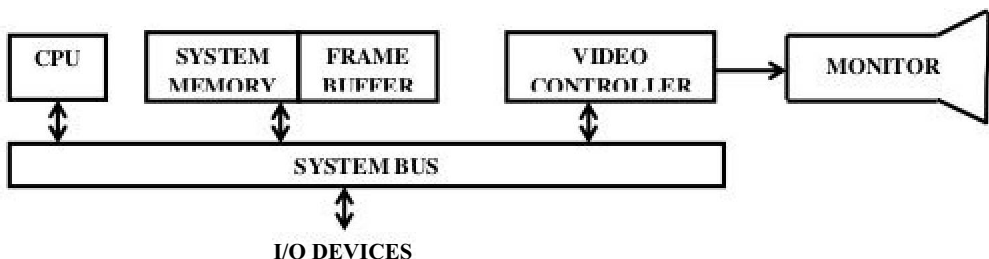
Raster-Scan Systems usually use several processing units. Besides the CPU (Central Processing Unit), a special processor called the video controller or display controller controls the display device. The frame buffer can be stored anywhere in system memory and is accessed by the video controller. Advanced raster systems may also use additional processors, called co-processors or accelerators, to perform graphics operations more efficiently.



VIDEO CONTROLLER:

A Video Controller, also known as a Video Display Controller (VDC) or Graphics Controller, is a hardware component responsible for controlling the display of text, images, and graphics on a computer monitor. It acts as an interface between the CPU, video memory (VRAM), and the display device.

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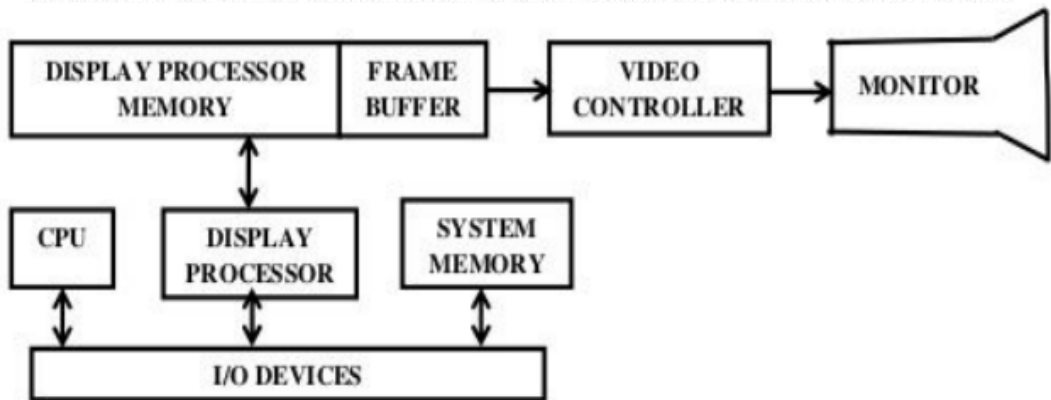


The CPU sends display data to the video controller, which stores it in video memory and continuously reads it to generate video signals for the monitor. It refreshes the screen many times per second to ensure smooth and flicker-free display. The video controller also manages important display parameters such as screen resolution, color depth, and refresh rate, allowing high-quality image output. Modern video controllers contain a Graphics Processing Unit (GPU) that accelerates the rendering of 2D and 3D graphics, reducing the workload on the CPU and improving overall system performance. Video controllers are widely used in computers, laptops, gaming systems, smartphones, and multimedia devices to provide fast, accurate, and high-quality graphical output.

RASTER-SCAN DISPLAY:

Raster- scan display is based on TV technology. In this system, the electron beam moves across the screen one row at a time from top to bottom. The image is stored in a memory area called the frame buffer (or refresh buffer), where each screen point is called a pixel (picture element). The frame buffer stores brightness and color information for every pixel, allowing realistic images with shading and color patterns. In a black-and-white system, a value of 1 means the electron beam is on, while 0 means it is off.

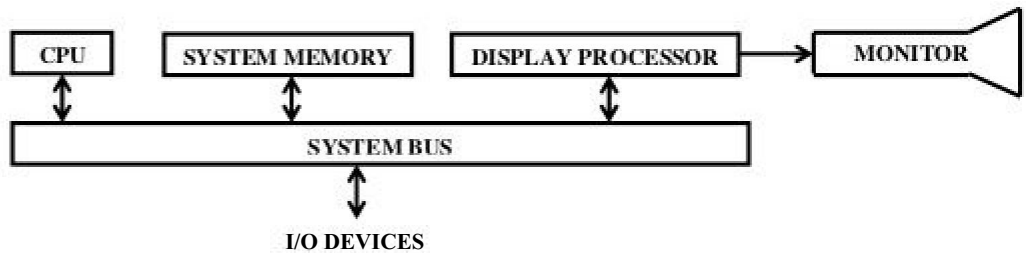
A frame buffer with one bit per pixel is called a bitmap, while one with multiple bits per pixel is called a pixmap. Raster-scan displays refresh the screen about 60–80 times per second (Hz). They usually include a graphics controller or display co-processor to reduce the workload of the CPU. One of its main tasks is scan conversion, which converts picture descriptions into pixel values and stores them in the frame buffer for display.



[RASTER SCAN SYSTEM WITH A DISPLAY PROCESSOR]

RANDOM-SCAN DISPLAY:

A Random-Scan Display system stores the application program and graphics data in the computer's memory. The picture information is kept in a display file, which is accessed by the display processor to refresh the screen. The display processor, also known as the Display Processing Unit (DPU) or Graphics Controller, controls the drawing of images on the screen. Instead of scanning the entire screen line by line, it draws only the required parts of the picture, making it suitable for displaying line drawings and vector graphics.



RANDOM-SCAN SYSTEMS:

A Random Scan System, also called a Vector Display or Calligraphic Display, is a display system in which the electron beam is directed only to the parts of the screen where the picture is to be drawn. It draws the image one line at a time instead of scanning the entire screen. The picture is stored as a set of line-drawing commands in a display list (refresh buffer). The refresh rate depends on the number of lines in the image. Random scan systems produce sharp and smooth line drawings and are mainly used in CAD, engineering drawings, and scientific applications. However, they cannot display realistic shaded images or photographs.

INPUT DEVICES:

The various input devices are: Keyboard, Mouse, Trackball and. Space-ball, Joysticks, Data glove, Digitizers, Image Scanner, Light Pens, Touch Panels and Voice System.

I. KEYBOARD:

It is used, primarily as a device for entering that strings. It is an efficient device for inputting non-graphics data as picture labels associated with a graphics display. It provided with features to facilitate entry of screen co- ordinates, menu selections or graphics functions.

Cursor-control keys and function keys are common feature on general purpose keyboards. Function keys allows users to entry frequently used operations in a single keystroke and a cursor control keys can be used to select displayed objects or co-ordinate positions by positioning the screen cursor.

II. MOUSE:

A mouse is a small hand-held box used to position the screen cursor. Wheels or rollers on the bottom of the mouse can be used to record die amount of direction of movement.

III. TRACKBALL AND SPACEBALL:

A track-ball is a ball that can be rotated with fingers or palm of the hand produce screen-cursor movement. It is a 2-dimensional positioning device. A space-ball provides 6 degrees of freedom. It does not actually move. It is used for 3-dimensional positioning.

IV. JOYSTICKS:

It consists of a small vertical layer mounted on a base that is used to steer the screen cursor around. Most joysticks select screen positions with actual stick movement, other respond to pressure on the stick. In movable joystick, the stick is used to activate switches that cause the screen cursor to move at a constant rate in the direction selected.

V. DATA GLOVE:

A data glove can be used to grasp a * virtual object'. It is constructed with a series of sensors that detect hand and finger motion.

VI. DIGITIZERS:

It is a common device used for drawing, painting or interactively selecting co-ordinate positions on an object. These devices can be used to input co-ordinate values in either a 2D or 3D space. It is used to scan over a drawing or object and to input a set of discrete coordinate positions, which can be joined with straight line segments to approximate die cute or surface shapes.

One type of digitizers is the "Graphics Tablet", which is used to input 2D co-ordinates by activating a hand cursor or stylus at selected, positions on a flat structure. A hand cursor contains cross hairs for sighting positions, while a stylus is a pencil shaped device that is pointed at positions on the tablet.

VII. IMAGE SCANNER;

Drawings, graphs, color and black and white photos or text can be stored for computer processing with an image scanner by passing an optical scanning mechanism over the information to be stored.

VIII. LIGHT PENS:

This is a pencil shaped device are used to select screen position by detecting the light coming from points on the CRT screen. An activated light pen, pointed at a spot on the screen as the electron beam lights up that spot generates an electrical plate that causes the co-ordinates position of die electron beam to be recorded.

IX. TOUCH PANELS:

It allows displayed objects or screen positions to be selected with the touch of a finger. A typical application of touch panels is for the selection of processing options that are represented with graphical icons.

Optical touch panels employ a line of infrared light emitting diodes (LEDs) along one vertical edge and along one horizontal edge contain light-detectors. These detectors are used to record which beams are interrupted when the panel is touched.

An electrical touch panel is constructed with 2 transparent plates separated by a small distance. One plate is coated, with conducting material and other with a resistance material.

X. VOICE SYSTEM:

It can be used to initiate graphics operations or to enter data. These systems operate by matching an input against a predefined dictionary of words and phrases.

HARD COPY OUTPUT DEVICES:

Hard-Copy output devices give images in several formats. For presentations or archiving, we can send files to devices or service bureaus that will produce 35mm slides or overhead transparencies. The quality of pictures obtained from a device depends on dot size and the number of dots per inch or lines per inch that can be displayed.

Printer produces output by either impact or non-impact methods. Impact printers press format character faces against an inked ribbon onto the paper. A line

printer is an example of impact device. Non-impact printers and plotters use laser techniques methods and electro-thermal methods to get images into paper.

Character impact printers have a dot matrix print head containing a rectangular array of protruding wire pins with the number of pins depending on the quality of the printer.

In a laser device, a laser beam creates a charge distribution on a rotating drum coated with a photoelectric material, such as selenium. Toner is applied to the drum and then transferred to paper. Ink-jet methods produce output by squirting ink in horizontal rows across a roll of paper wrapped on a drum. The electrically charged ink stream is deflected by an electric field to produce dot-matrix patterns.

An electrostatic device places a negative charge on the paper, one complete row at a time along the length of the paper. Then the paper is exposed to a toner. The toner is positively charged and so is attracted to the negatively charged areas, where it adheres to produce the specified output. Electro-thermal methods use heat in a dot-matrix print head to output patterns on heat-sensitive paper.

GRAPHICS SOFTWARE:

Graphics Software is of 2 types. That is:

- i) General Programming Packages
- ii) Special Purpose Application Packages.

A general programming package provides an extensive set of graphics functions that can be used in a high level programming language. Application graphics packages are designed for non-programmers. So, that user can generate displays without worrying about how graphics operations.

CO-ORDINATE REPRESENTATIONS:

With few exceptions, general graphics packages are designed to be used, with Cartesian co-ordinate specifications. If co-ordinates values for a picture are specified in some other references frame, they must be converted to Cartesian co-ordinates before they can be input to the graphics package.

Special purpose packages may allow using of other co-ordinate frames that are appropriate to the application. We can construct the shape of individual objects, in a scene within separate co-ordinate reference frames, called "modeling co-ordinates" or "local co-ordinates" or "master co-ordinates".

Only individual object shapes have been specified, we can place the objects into appropriate positions within the scene using a reference frame called world co-ordinates. The world co-ordinates description of the scene is transferred to one or more output device reference frames for display. The display co-ordinate systems are referred to as device co-ordinates/screen co-ordinates in the case of video monitors.

Generally; a graphics system first converts world co-ordinates to specific device co-ordinate to specific device co-ordinates. An initial modeling co-ordinate position (X_{jtt}, y_{inc}) is transferred to advice co-ordinate position $(x^{\wedge}, y^{\wedge})$ with the sequence (x_{Jlx}, y_{Jlx}) (x_{wc}, y_{wc}) (x_{llc}, y_{llc}) $(x_{dt}, y^{\ast})$. The normalized co-ordinates satisfy the inequalities $0 < x_{nw} < 1, 0 < y_{llv} \leq 1$ and the device co-ordinates $^{\ast}x_{dc}$ and by_d are integers within the range $(0,0)$ to (x_{inaK}, y_{llak}) for particular device.

GRAPHICS FUNCTIONS:

A general purpose graphics package provides users with a variety of functions for creating and manipulating pictures. The basic building blocks for pictures are referred to as output primitives. They include character strings and geometric entities such as points, straight lines, curved lines, filled, areas and shapes defined with arrays of color points.

Attributes are the properties of the output primitives that is an attribute describe how a particular primitive is to be displayed. We can change the size, position or orientation of an object within a scene using geometric transformations similar modeling transformations are used to construct a scene using object description given in modeling co-ordinates.

Viewing transformations are used to specify the view that is to be presented and the portion of the output display area that is to be used. Pictures can be subdivided into component parts called structure s/segments/objects, depending on the software package in use.

Interactive graphics applications use various kinds of input devices, such as a mouse, a tablet or a joystick. Input functions are used to control and process the data flow from those interactive devices. A graphics package contains a number of housekeeping tasks, such as clearing a display screen and initializing parameters. The functions for carrying out these chores under the heading control operations.

SOFTWARE STANDARDS:

The primary goal of standardized graphics software is probability. When packages are designed with standard graphics functions, software can be moved

easily from one hardware system to another and used in different implementations and applications.

International and National standards planning organization have co-operated in an effort to develop a generally accepted standard for computer graphics. After considerable effort, this work on standards led to the development of the Graphical Kernel System (GKS). This system was adopted as the first graphics software standard by International Standard Organization (ISO), and by others.

The 2nd software standard to be developed, and approved by the standards organization was Programmer's Interactive Graphics Standard (PUIGS), which is an extension of GKS.

Standard Graphics functions are defined as a set of specifications that is independent of any programming language. A language binding is then defined for a particular high level programming language. Standardization for device interface methods is given in Computer Graphics Interface (CGI) system and the Computer Graphics Metafile (CGM) system specifies standards for archiving and transporting pictures.

PUIGS WORKSTATION:

The workstation is a computer system with a combination of input and output device that defined, for a single user. In PUIGS and GKS, the term workstation is used to identify various combinations of graphics hardware and software.

A PUIGS workstation can be a single output device, single input device, a combination of input and output devices, a file or even a window displayed, on a video monitor. To define and use various "workstations" within an application program, we need, to specify a workstation identifier and the workstation type.